Single-core CPU: DRAM preprocessing + inference

Created 2026-11-09 · Measurements 2026-11-09

Host hacc-build-03: Intel Xeon Gold 6234, CPU2 on NUMA2, SMT sibling CPU34; 32 KiB L1d, 1 MiB L2 per physical core and 24.75 MiB shared L3 per socket. All worker threads are restricted to CPU2. The raw bitstream begins in local DRAM and is evicted from CPU data caches before primary calls. Compiled code/model are warmed. The complete pipeline is measured through materialized CPU verdict.

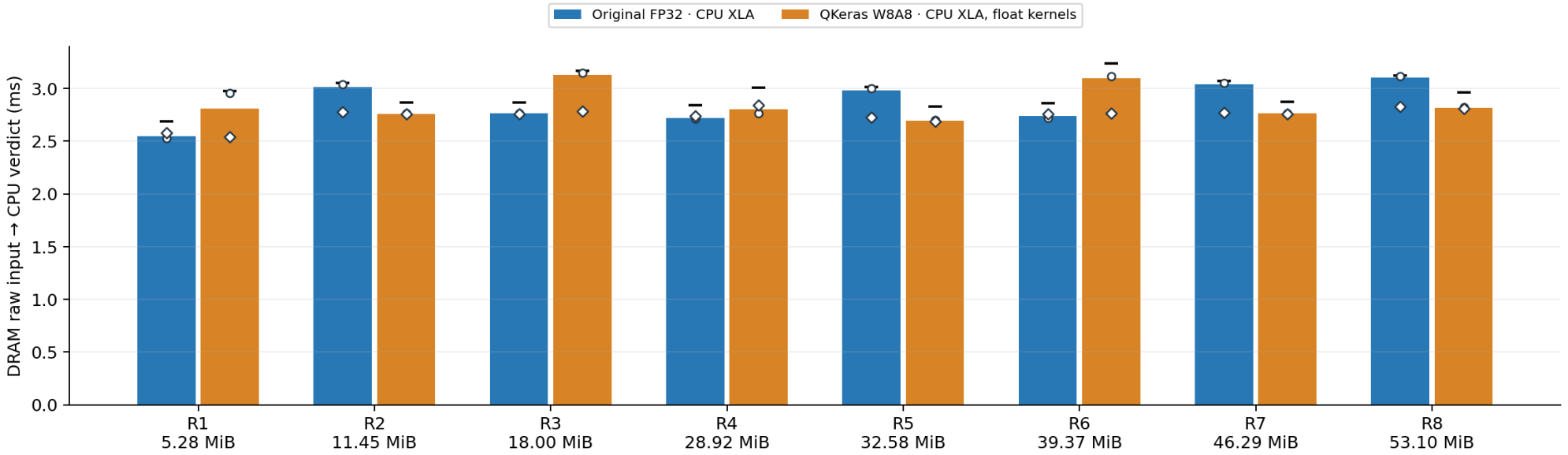
19,200 primary calls across two models, two input-cache scopes and twelve workloads; **33,600 measured calls** including separate stage diagnostics. Each cell has 400 measurements from two independently launched rounds, after 20 warmups per round.

Repeatability limitation: cold-input medians differ by up to 14.1% between independent process sessions. The figures expose both sessions; pooled medians should not be interpreted as a uniquely stable machine latency.

Input	MiB	FP32 cold (ms)	QKeras W8A8 cold (ms)
R1	5.28	2.547	2.809
R2	11.45	3.016	2.761
R3	18.00	2.763	3.131
R4	28.92	2.722	2.804
R5	32.58	2.981	2.694
R6	39.37	2.739	3.096
R7	46.29	3.041	2.762
R8	53.10	3.103	2.815

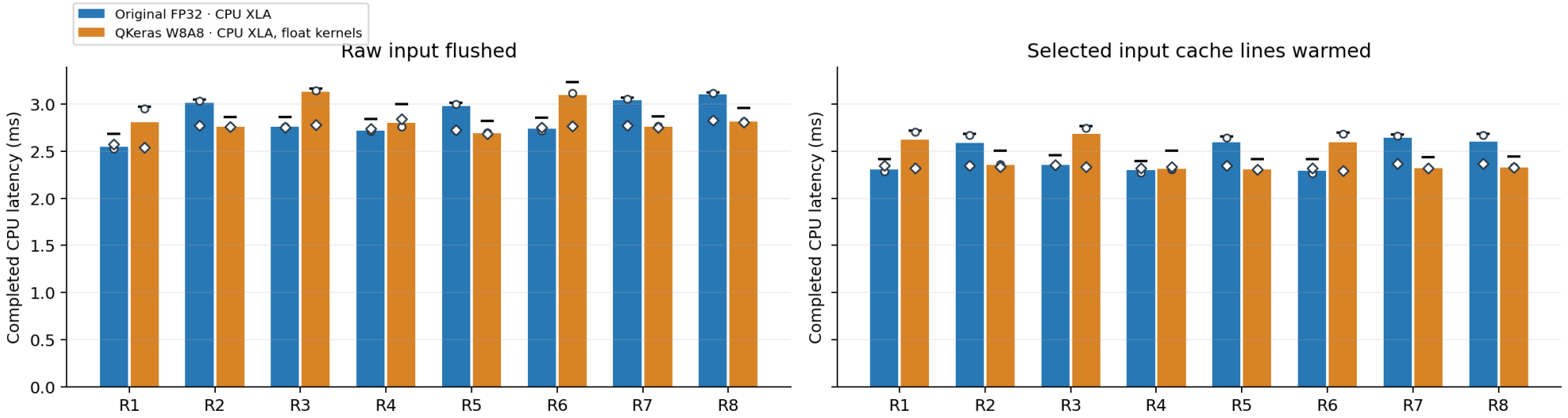
The QKeras result measures the frozen quantized model using floating CPU operators. It does not establish the performance of native INT8 instructions. Primary models and preprocessing are validated against their own CPU references; no equality of FP32 and quantized scores is assumed.

1 · Single-core CPU verdict latency: cold raw input



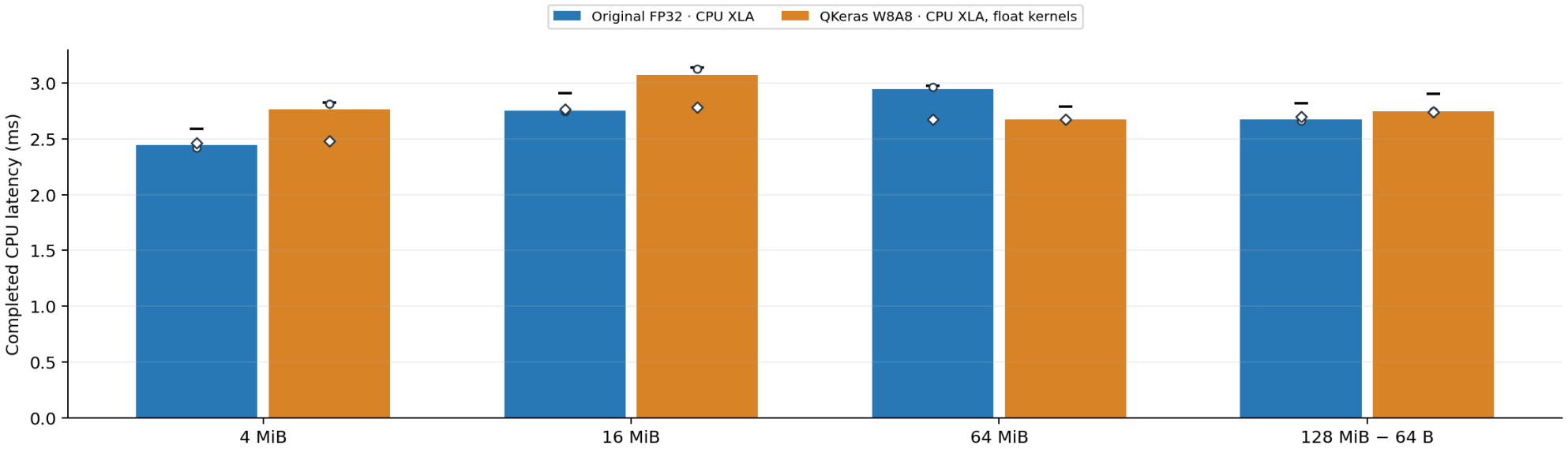
Raw input is explicitly flushed from CPU data caches before each call; model/code are warmed. Bars are pooled medians, black ticks p95, white circles round A medians and diamonds round B medians. Each workload/model contributes 400 measured calls, equally split across rounds after 20 warmups. Both use CPU XLA + cached ConcreteFunction; QKeras W8A8 uses quantized values with floating CPU operators, not native INT8 kernels.

2 · Cache-state control: cold versus warmed selected input lines



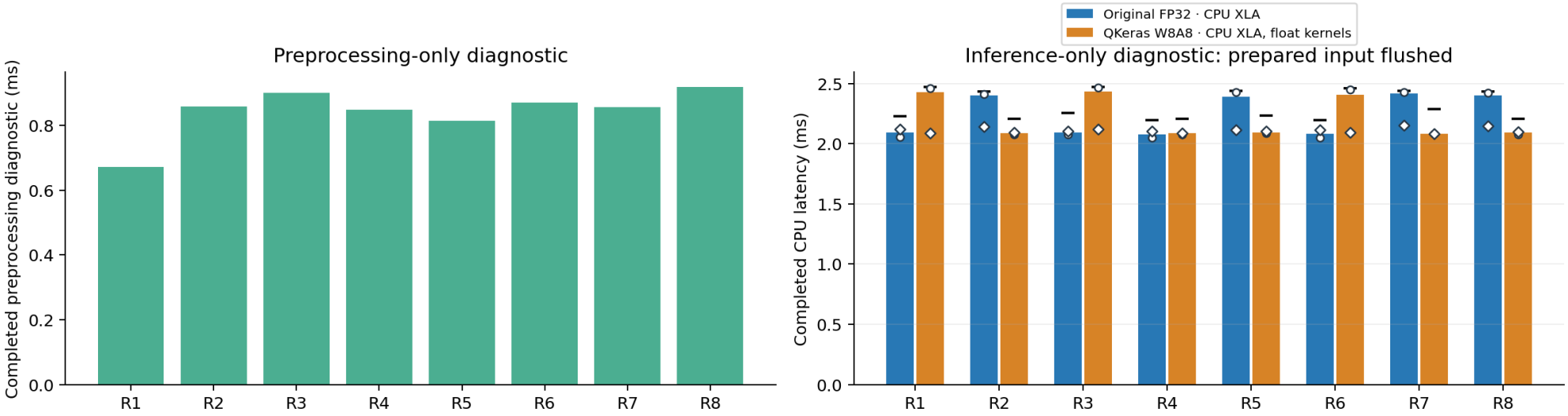
The warm control touches only the source bytes selected by preprocessing. Above the sampling cutoff these occupy approximately 4 MiB of cache lines, which fits the socket's 24.75 MiB L3; the full 128 MiB buffer does not. Cache preparation is untimed. The cold condition invalidates the actual aliased input storage. Natural cache fills and prefetching during execution remain enabled; this is not cache-bypassing inference.

3 · Established synthetic workloads: cold-input verdict latency



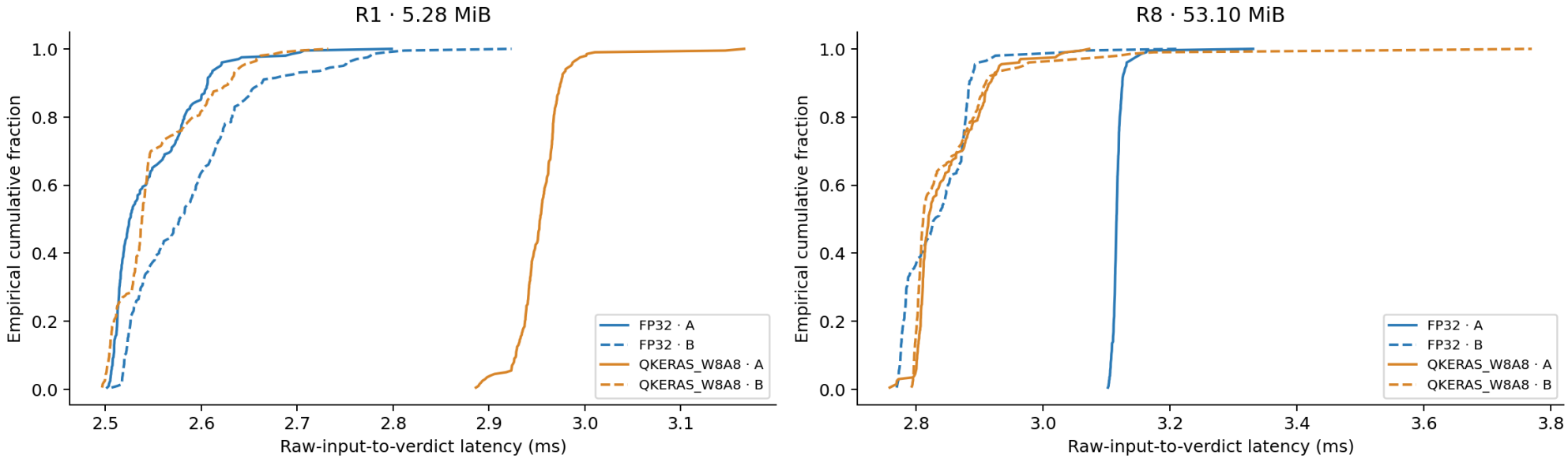
Pinned synthetic byte identities match the previous HBM campaign. Each bar uses 400 measured calls. Input lengths affect which bytes are gathered, memory locality and compiled graph shape. Sampling touches 65,536 byte positions rather than casting or scanning the complete raw file into floating point.

4 · Separate preprocessing and inference diagnostics



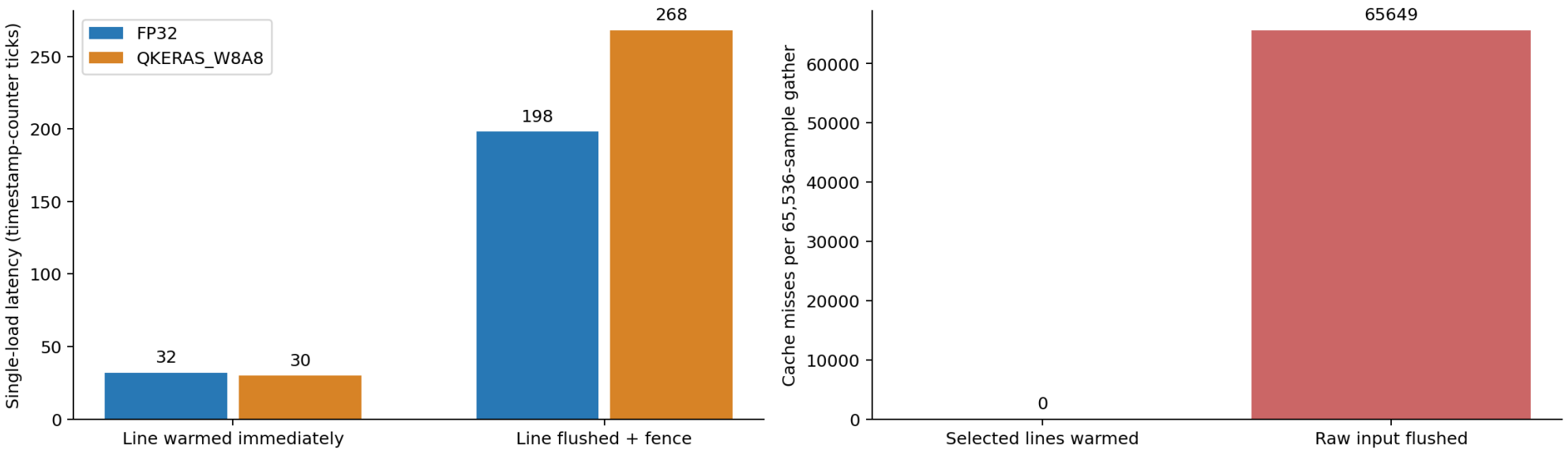
These are independent completed-call timings, not an additive breakdown of the fused end-to-end graph. Preprocessing-only returns/materializes a 256 KiB float32 tensor; fused inference does not expose that intermediate to Python. Inference-only starts from the prepared tensor flushed from data caches. Preprocessing is shared by the two primary models: integer sampling/padding followed by an exact 256-entry float32 normalization lookup. Boundary validation caught and rejected a prior rounding-sensitive arithmetic implementation.

5 · Latency distributions retain independent process sessions



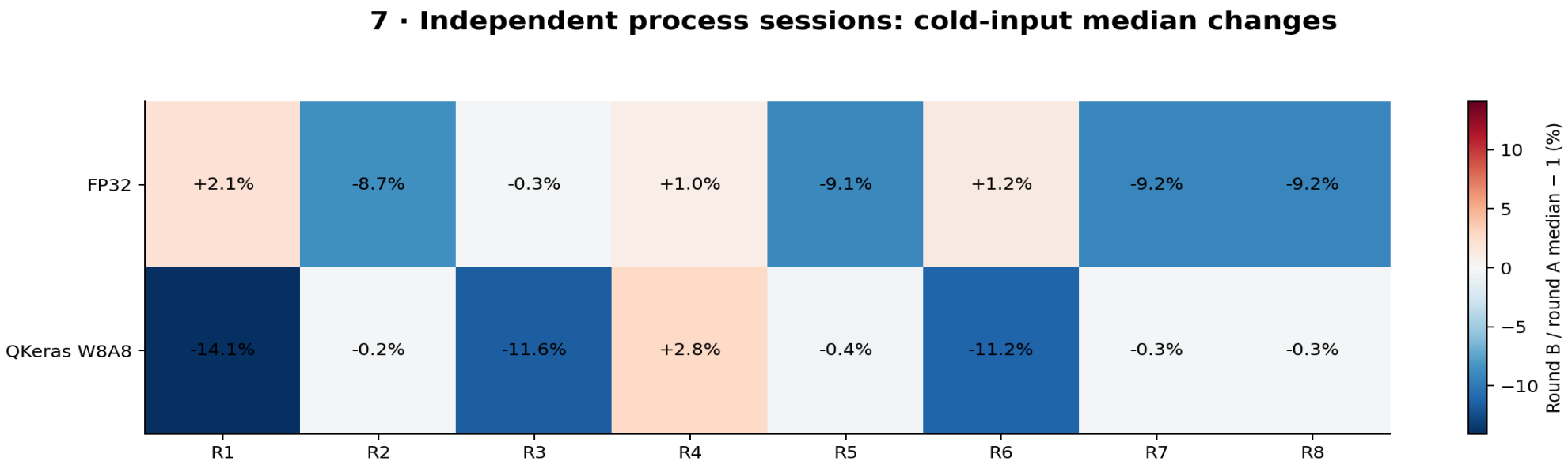
Two independently launched, model-warmed rounds use reversed model, workload and cache-scope order. All accepted calls remain present; latency variation alone does not trigger selective reruns. This is a shared build host: affinity limits our worker to one logical CPU, while another process could still contend for the core, SMT sibling or shared resources. Environment/load and process CPU-time evidence are retained.

6 · Cache eviction evidence from isolated CPU diagnostics



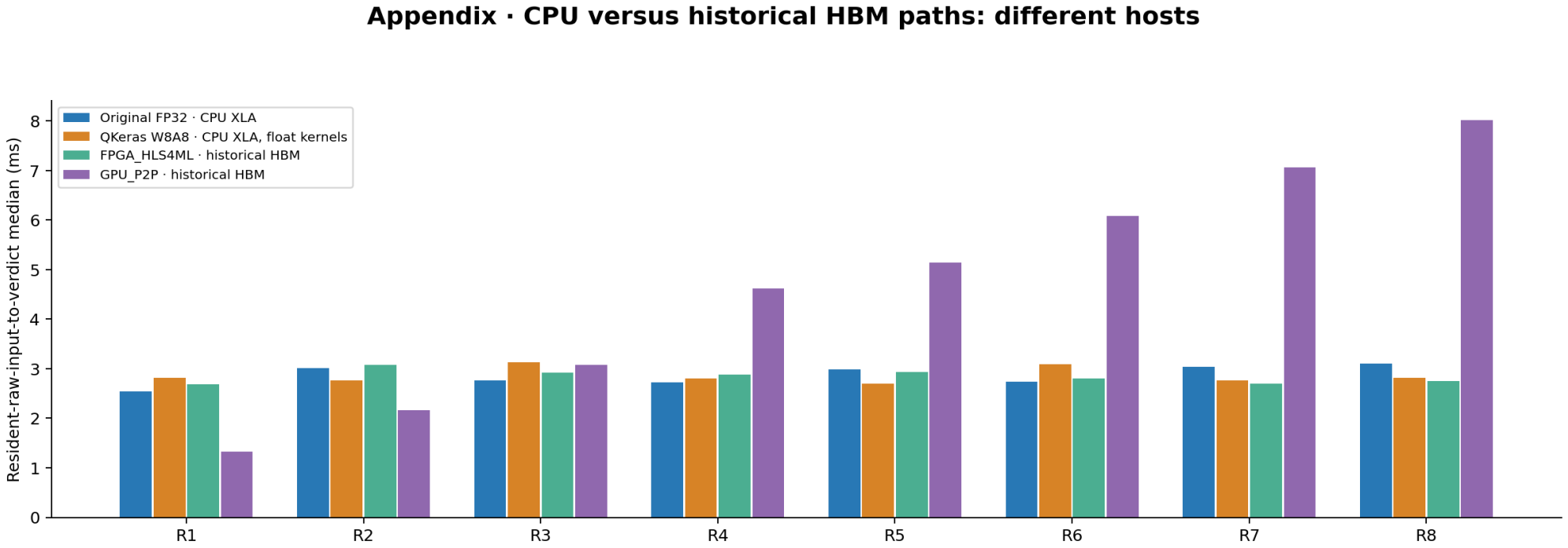
Left: 512 randomized line probes per cache state; fences and timestamp-counter overhead are included. Timestamp-counter ticks are not actual variable-frequency core cycles. Right, when available: per-thread hardware cache-miss counters enabled only around a native gather on a 128 MiB–64 B arena, after untimed cache preparation. This diagnostic is separate from TensorFlow/model timings. CLFLUSHOPT + fences invalidate input lines before a call; they do not prevent normal cache use during the call.

7 · Independent process sessions: cold-input median changes



Round B reverses model, workload and cache-scope order. Session changes are retained rather than rerun away; separate absolute values appear in the appendix. CPU frequency snapshots vary and this is a shared host. No single physical cause for these differences is established. Equal 200-call sessions contribute to each pooled 400-call primary cell.

Appendix · CPU versus historical HBM paths: different hosts



CPU: Xeon Gold 6234, hacc-build-03, one logical CPU, local DRAM, measurements 2026-11-09. FPGA/GPU: original HBM report, measured 2026-11-08 on hacc-gpu-u55c-01 with EPYC host, U55C device2 and MI210 GPU0. Accelerator values are FPGA accepted-arm→verdict events; CPU uses completed host wall time. These are different-machine deployment-path observations, not a controlled same-host speedup claim. Historical OPT_10 changed by up to 9.825% after identical-image reprogramming; this plot uses the report's initial-program medians. QKeras CPU and hls4ml FPGA also differ in fixed-point accumulator/input arithmetic; FP32 uses the same pinned float checkpoint as GPU.

Appendix · Per-session cold-input medians

Each cell is A / B in milliseconds; 200 measured calls per process session. Main medians pool equally sized sessions.

Model	R1	R2	R3	R4	R5	R6	R7	R8
FP32	2.526 / 2.578	3.038 / 2.775	2.764 / 2.756	2.715 / 2.742	2.999 / 2.726	2.722 / 2.756	3.054 / 2.774	3.116 / 2.830
QKERAS_W8A8	2.954 / 2.538	2.764 / 2.758	3.146 / 2.782	2.764 / 2.842	2.697 / 2.688	3.117 / 2.767	2.765 / 2.757	2.820 / 2.811

Model identities and interpretation

FP32 uses the original unquantized Keras checkpoint from the accepted GPU R06 runtime. QKeras W8A8 uses the frozen OPT_6 architecture and pruned/QAT weights. It preserves quantization functions and 8-bit values through floating CPU operators. It must not be cited as a native INT8 CPU implementation. The hls4ml software simulator is not used as a CPU performance baseline. SIMD/vectorization on one CPU core is allowed.

Native INT8 conversion investigation

A separately labeled native INT8 conversion was attempted: status **diagnostic_only**. Conversion changes tie rounding and integer pooling/output scales; it is not bit-exact QKeras or hls4ml. All 26 admission verdicts match, but its calibrated output range clips several original logits (maximum score difference 18.237). It is not accepted as a faithful replacement or primary timing baseline. No native INT8 speedup is claimed. Its calibration/provenance, rejected disposition and score differences are retained under campaign int8_conversion/. Calibration was not retuned on benchmark outputs.

Reproducibility and measurement limits

The campaign retains source/model/input hashes, input alias tests, 26-case admission, raw trials, thread affinities, local NUMA placement, page backing, cache setup times, CPU frequency/load snapshots and numerical/visual review. File reads, DRAM preload, page faults during initial setup, model loading, graph compilation, cache preparation and warmups are excluded from primary timing. Runtime materialization and thresholding of the final scalar verdict are included. Code/model are warmed; intermediate activations naturally use caches. TLBs and instruction caches are not explicitly flushed. No global cache dropping, governor changes, FPGA programming or GPU execution occurred.